NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO

FACULTY OF ENGINEERING

Computer Graphics and Human-Computer Interaction

3D graphical simulation of Universidad station on Line 3 of the Mexico City Metro

Group: 02

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1. Abstract

This project consists of developing a three-dimensional, interactive graphical simulation of Universidad station on Line 3 of the Metro Collective Transportation System. The purpose of the project is to represent its main spaces, architectural elements, and characteristics inherent to the operation of a system of this type. To develop it, we used Blender to create and organize geometric models, which were later exported in .fbx format and integrated with the C++ language using the OpenGL API. This simulation has 3 different scenes: the upper part of the station, the platform, and the interior of a subway car. This simulation allows the user to take virtual tours using different camera modes, as well as switch between scenes through control points activated by keys to move between them. The program also allows the user to manipulate objects, as explained in greater detail in this project's manual. In addition to these actions that the user can perform, the program itself implements animations of a metro train and all the elements inside it, the opening of doors and turnstiles, the movement of characters, materials with different properties and colors, different textures, and lighting techniques such as Phong and Fresnel. In this way, the project integrates the concepts of modeling, rendering, and animation, in addition to the key aspect of human-computer interaction to virtually represent infrastructure such as public transportation.

2. Introduction

2.1 Context and problem

The accelerated growth of urbanization has made mobility one of the main challenges of sustainable development. More than half of the world's population lives in urban areas, and this is estimated to reach nearly 70% by the middle of the 21st century.

These areas account for between 60% and 80% of energy consumption and approximately 75% of carbon emissions, and in 2022 only half of their inhabitants had adequate access to public transportation (United Nations, 2026). Therefore, Target 11.2 of Sustainable Development Goal 11 seeks to facilitate access to safe, affordable, and sustainable transportation systems (United Nations, 2026).

In Mexico City, the Metro transports more than three million people a day (Campuzano, 2024). Universidad station, the southern terminus of Line 3, recorded an average of 60,918 users per working day in 2024, ranking fifth among the busiest stations (Campuzano, 2024).

Line 3 experiences heavy crowds: in June 2026, waits of up to ten minutes and overcrowded trains were reported, and the Metro responded by dispatching empty trains to speed up service

(Publimetro, 2026). Added to this is its age: the current trains on the line are more than 40 years old and have traveled more than 3 million kilometers.

For this reason, the Government of Mexico City announced a renovation with an investment exceeding 40 billion pesos and the purchase of 45 new trains (Infobae, 2026). In addition, Universidad station will become a transportation hub by connecting with the future Line 4 of Cablebús and Line 14 of Trolebús (Chilango, 2026). This makes it a representative station of growing importance to study.

2.2 State of the art

Railway simulators. Among commercial simulators, Train Sim World released its seventh installment with new routes and trains (Dovetail Games, 2026), while OpenBVE is a free, open-source simulator (The OpenBVE Project, n.d.).

In both cases, the experience revolves around driving the train and the fidelity of routes and rolling stock. This project adopts a different point of view: that of the user who enters the station, finds their way, and boards the car.

Simulation and virtual reality in metro stations. In academia, virtual environments are used to study how people find their way inside a station. Wei et al. (2025) built a virtual station and, with 60 participants, found that hanging signs improve wayfinding the most, followed by those on walls and columns and, lastly, those on the floor.

Yu et al. (2023) recreated a large transfer station with its actual signage to analyze how people pay attention to signs and build spatial knowledge under normal conditions and pressure. Hu and Xu (2023) compared vertical, horizontal, and mixed signage in a Nanjing station with 36 volunteers. Hu (2017) proposed a 3D simulation method to evaluate passenger flow and facility layout at stations.

The case studies reported by Wei et al. (2025) and Hu and Xu (2023) concern Chinese cities, and in our search we did not find equivalent studies of Mexico City Metro stations.

Graphics technology. The system is developed in C++ with OpenGL 3.3, whose specification defines the programmable pipeline through shaders (Segal & Akeley, 2010). This allows direct control over geometry, lighting, texturing, and hierarchical animation, rather than leaving them to a game engine.

2.3 Economic, social, and ethical analysis

Economic. The project has a low cost because the modeling and development tools are open source, and Visual Studio in its Community version is free. The context being represented is large in scale: the renovation of Line 3 will exceed 40 billion pesos, and the government committed to keeping the ticket price at five pesos (Infobae, 2026). In this context, an interactive simulation is a low-cost way to observe how the spaces of a station are organized.

Social. Target 11.2 calls for attention to those who depend most on public transportation. For this reason, the scene includes accessibility elements (elevators, signage, exclusive areas) and adult and child characters. The simulation also serves as an educational tool: it allows someone who has never used the station to recognize its spaces and routes without the risk of getting lost, and it makes visible parts of the infrastructure that normally go unnoticed. It also helps tourists feel encouraged to use public transportation when they visit Mexico City and, together, helps reduce pollution and car use per family/person.

Ethical. Four aspects were addressed: (1) authorship: at least 80% of the models are original creations of the team, and those from third parties or generated with AI are referenced; (2) responsible use of AI: it is declared, cited, and the prompts are attached; (3) representation: the STC brands and signage are used for educational purposes, and the system does not replace official information; (4) privacy: the characters are generic and do not represent real people.

2.4 Purpose and scope

Based on the above, this project presents an interactive graphical simulation of Universidad station on Line 3 of the Mexico City Metro, with 86 geometric models, 4 camera modes, lighting, cube mapping (based on those proposed by the professor), and basic keyframe animations. The team decided to limit the scope to a single station in order to focus on modeling quality. The rest of the report presents the methodology, experiments, results, and individual conclusions.

Note on AI use: Claude Sonnet 5.5 (Anthropic, 2026) was used in this Introduction to improve the wording. The team read and modified the text (see Appendix A).

3. Practical methodology

3.1 Architecture

The development of the project was divided into different stages, including observation, analysis, implementation, and testing. Collaborative work among team members was necessary across different activities so that everyone could support the completion of the entire project.

To begin the project, we started the stage of creating the three-dimensional models, applying textures when necessary. We proposed creating 3 scenes to integrate all our models in an organized way and then implement the programming of the graphical environment.

Blender 5.2.2 was used to create the models, and Visual Studio and OpenGL were used for interactive development. We also used the GitHub version control system to support proper collaborative creation and keep development organized.

3.2 Analysis and planning of the virtual environment

To begin developing our virtual environment of Universidad station in Mexico City, it was necessary to visit the station—an activity carried out by different team members—to identify and capture the different elements or objects present at the station. Several photos and videos were taken so as not to omit important objects at the station. We also used as a reference an online video showing a tour of the renovated Line 3 of the Mexico City Metro.

A document was created to list the models to be developed, including signage elements, elements found inside the train, structural elements, and elements found inside the station. During one of the visits to the station, reference measurements were taken for the creation of the models.

To create the virtual environment, all models were divided into three scenes: the station entrance, the platform area, and the train. This division helped us develop the final models independently and then begin programming the graphical environment.

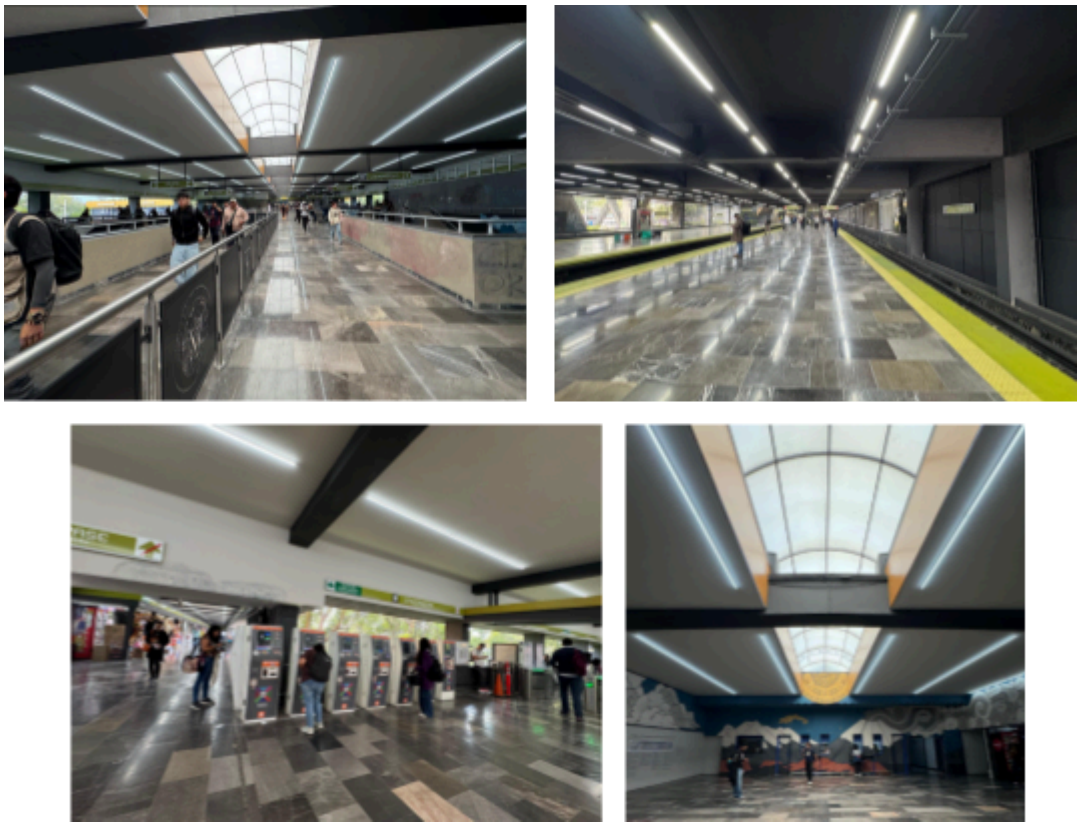


Figure 1. Reference images taken during the visits.

3.3 Design and organization of scenes

To successfully create a virtual environment of the Universidad Metro station, three scenes were designed: the station entrance, the platforms, and the complete train. Each scene includes different elements and characteristics of its own. In addition, for the cubemap we have images provided by the professor; we used the same structure.

The first scene mainly includes the upper part of the station, with elements such as turnstiles, recharge machines, sales stalls, etc. The second scene is the platform area, which has signage, stairs, tracks, etc. The third scene is the complete train, which has doors, windows, seats, handrails, etc.

Scene	Objects
Station entrance	"Elevator" sign, "Platforms" sign, "Do not enter" sign, "Exit" sign, turnstile base, turnstile doors, card reader, metal barriers, turnstile indicator light, card recharge machine, candy stand (cylinder), candy stand (cube), ticket booths, ticket-booth glass, station manager's room, Exit 1 mural, metal mural protection, metal mural plaque, ticket-office mural, cybercenter room, cybercenter and station manager's door, computer, computer desk, attendant's desk, cybercenter and station manager's windows, steel plaque with coats of arms, station metal doors, bicycle parking

Platform	Station (corridor), fire extinguisher, station name sign, "Indios Verdes" direction sign, women-only area sign, "Terminal Universidad" sign, "Next departure" sign, "next departure" alarm, "Evacuation route" sign, "No smoking" sign, fixed stairs, stair railing, elevator, security cameras, gravel, track signal, steel rails, exclusive-area cone, steel fence, wooden planks, platform tile floor, station lamps, structural column, exterior platform column, structural beam, dome, yellow floor strip, stairlift
Train	Car (body), metro door, door glass, upper window, lower window, window frame, front windshield, seats, metal seat base, handrails, VIP handrail, interior door handle, lamp cover, fan, Metro lines map, Line 3 map, "emergency lever" sign, emergency lever, emergency lever box, "reserved seat" sign, cab (body), speakers, train wheel, car coupling piece

Figure 1. Division of the 3 scenes and their elements.

For the creation of the complete models (the 3 scenes), collections, empties, and parent-child relationships (Ctrl P -> Keep Transformations) were used; if a parent object moved, its child moved as well, allowing us to group several models belonging to a single scene into one *.blend* file. This organization allowed us to maintain an ordered, hierarchical structure for later implementation in OpenGL.

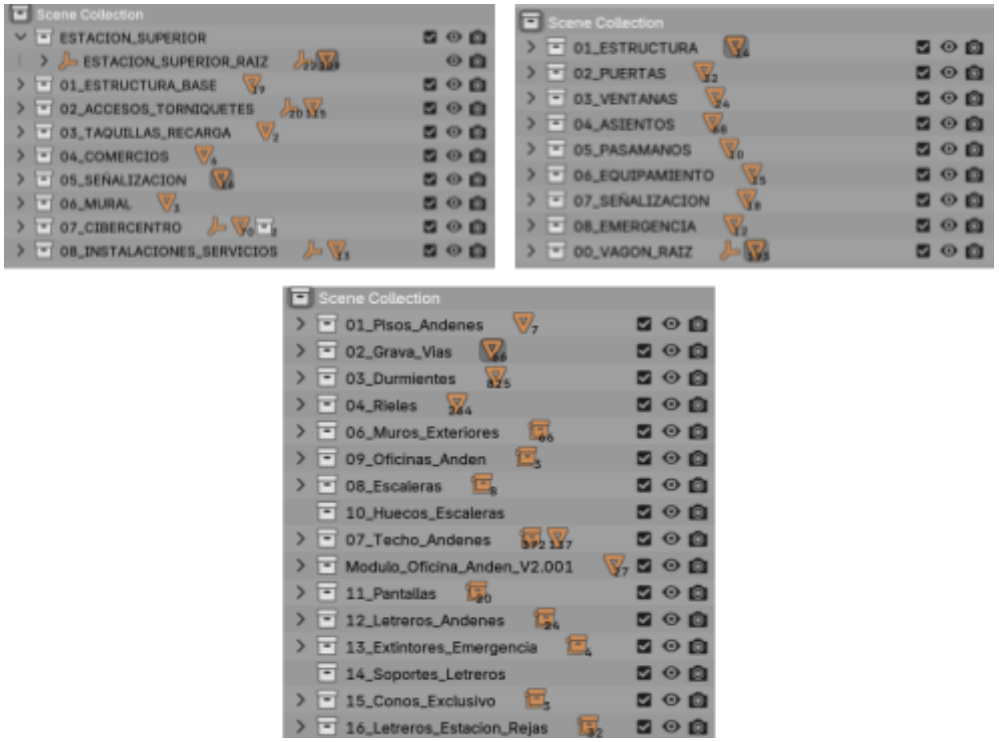


Figure 2. Collections implemented in the different scenes to apply hierarchies and parent-child relationships.

3.4 Geometric and hierarchical modeling

Blender version 5.2.2 was used to create the models. To model different elements, we started with basic geometric shapes, especially cubes and cylinders, which took shape by editing their dimensions, rotation, extrusions, etc.

To add detail to the models and make them somewhat more realistic, various modifiers were used, such as Bevel to soften shapes; Array to duplicate objects; Mirror to reproduce exactly what had been created, but mirrored; and Subdivision Surface to edit the object's base structure. All models were created using a single scale so that it would be easier to arrange them when assembling the complete models, although scaling was still performed on the final objects.

For hierarchical modeling, the different objects were organized into separate collections, with some, as in the case of the metro, having their own Empty for each car. Structuring the models in this way allowed us to manipulate compound elements without using Join, since some elements were dynamic. In addition, the use of Empty allowed us to rotate, translate, and scale compound elements instead of applying transformations one by one.

After completing the modeling and collection-merging work, each dynamic object was reviewed individually to ensure that it did not lose this property, before exporting it as *.fbx* and adding it to the development in OpenGL.

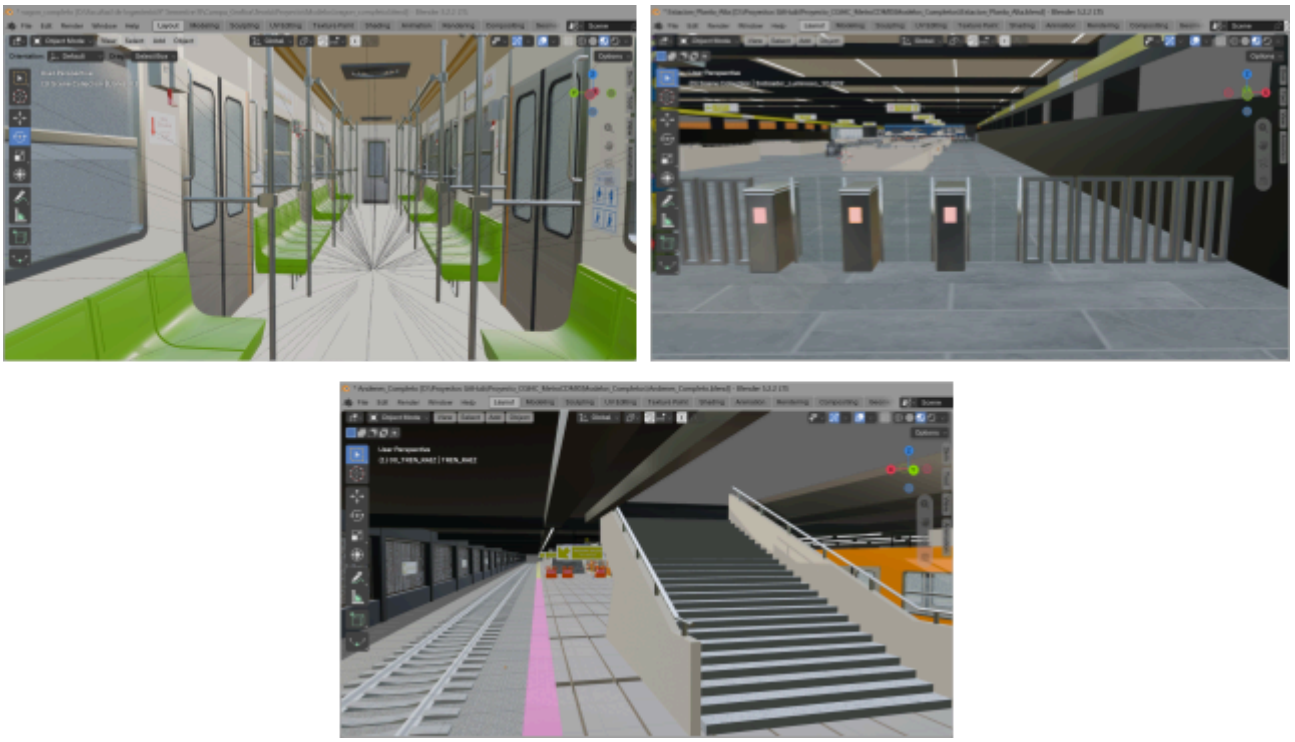


Figure 3. Application of Array to metro seats, station barriers, and stair steps.

3.5 Materials, colors, and texturing

To give our objects better visibility and appearance, different materials were created to represent their characteristics. In Blender, we applied the base color to each material so that, when implementing it in OpenGL, we could add the lighting elements.

The types of materials we used, in general, are translucent, matte, metal, and plastic. However, some materials were not easy to represent with base colors and lighting, so textures were applied using the UV Editing workspace in Blender, allowing OpenGL to use the texture coordinates. This procedure enabled us to add images mainly to our signage objects.

Likewise, to be able to access the objects created by team members, it was necessary to save them in a special way to transfer the resources; we used: File - External Data - Pack Resources. To export the materials to OpenGL without having to create a folder with textures, a script was created with the help of ChatGPT to recover the images used as textures.

3.6 Architecture and implementation in OpenGL

After creating the complete models, the next step was to implement them in OpenGL using the Visual Studio development environment. The complete models were exported in .fbx format; before that, all transformations were applied to preserve their dimensions, orientation, and scale so they could be added to the program.

The Assimp library was used to import the model and display its textures and colors. To keep the code development organized, folders were created for the complete models, the images used in the textures, and the shaders.

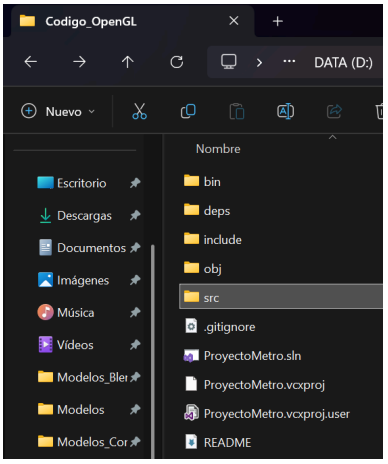


Figure 4. Organization of the code to be implemented in OpenGL

GLSL shaders were used for processing and rendering; they allowed us to perform geometric transformations and visual effects. In this way, we were able to visualize our 3 different scenes representing the complete station and the metro.

3.7 Lighting and shading

To give the objects a more realistic appearance, Phong local lighting models were applied. This model calculates how light reaches objects, and is divided into three:

1. Ambient: Simulates the general and indirect lighting of the space

   In our project, it is applied as a base light throughout the station and inside the cars. This is because it simulates indirect light bouncing off the walls, ceiling, and floor. In an enclosed space such as the metro, without ambient light, areas not directly lit by a lamp would appear as black shadows. The image below shows this type of lighting.

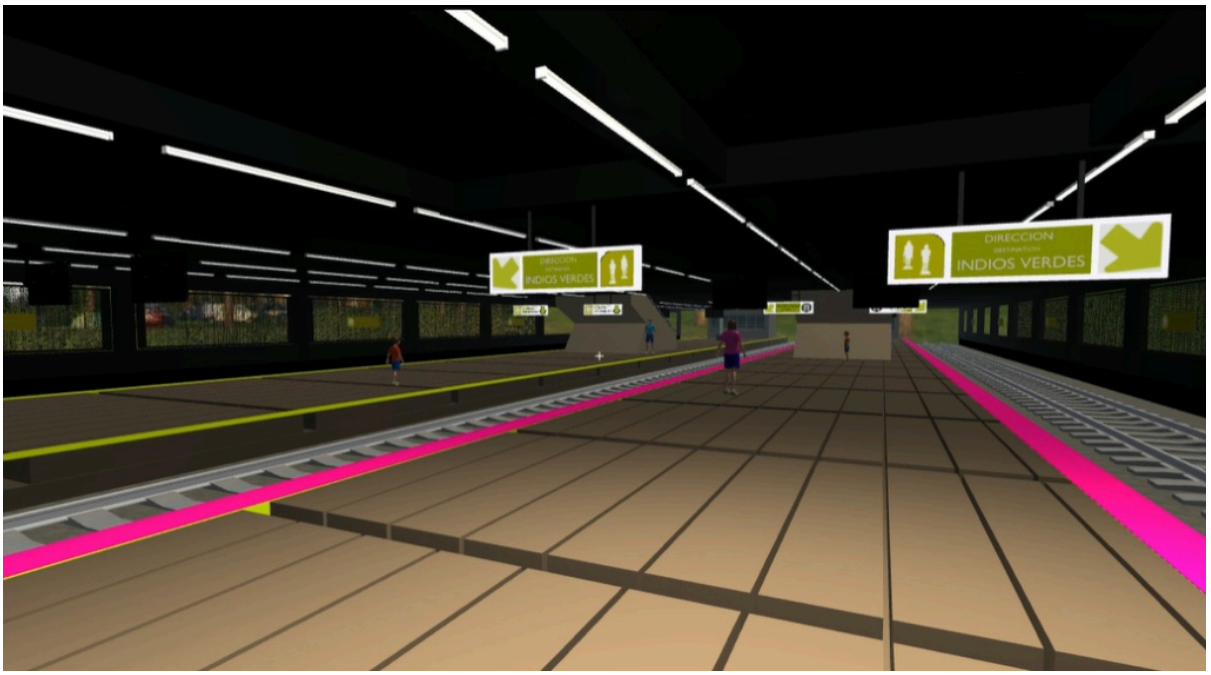


Figure 5. Ambient lighting in the platform scene

2. Diffuse: Creates reflected highlights.

   In our project, this type can be seen on the handrails, the stripes and metallic finishes of the train, the glass/window surfaces of the cars, the screens, and the tiled floor. Diffuse lighting is important for showing the reflective texture of the station's metallic materials, distinguishing them from matte finishes. Below is an image of this lighting on the orange train.



Figure 6. Diffuse lighting in the train scene

3. Specular: Defines the color and orientation of surfaces facing the lamps

   In our project, this type can be seen from the tubular lamps on the ceilings of the platforms and in the upper part. It also makes it possible to appreciate the characteristic texture and tone of the tiled floor on the platforms and the walls of the vestibule.



Figure 7.. Diffuse lighting in the upper part of the station.

3.8 Animations and interactions

To animate objects accurately, the keyframes.h file was used. It works as follows:

3.8.1 Keyframe Animation

- A "track" is defined and given a list of keyframes. Each keyframe has a specific time and a value (which can be a position, a rotation, or a camera vector), along with a smoothing curve.
- Smoothing (Easing): To keep movements from feeling robotic, the system includes functions such as IN, OUT, or IN_OUT. This makes the train or camera start slowly, accelerate to full speed, and brake smoothly when reaching its destination.

3.8.2 Interactions and First-Person Movement

User interaction with the environment was handled by combining free movement and physical restrictions:

- Camera Control (camera.h): The user can walk using the keyboard (FORWARD, BACKWARD, LEFT, RIGHT) and look around by moving the mouse. The camera calculates the Yaw and Pitch angles (preventing the screen from turning upside down) to generate the view matrix (LookAt) in real time.
- Collision System (colisiones.h): To prevent the person from passing through walls, desks, or stairs, walkable zones (rectangles on the plane) and box-shaped obstacles (AABB) were defined in the code. If the user tries to collide with something, the system stops forward movement to keep them within the permitted platforms and corridors.

3.9 Algorithms and flowcharts

The following collision, interpolation, and hierarchical-structure algorithms were developed and implemented in writing the code.

3.9.1 Collision Algorithm

This algorithm prevents the user from colliding with a corner or diagonal wall.

1. The program receives the player's current position and the desired position to which they want to move.
2. It first attempts movement along the X axis. If the new position is clear of obstacles and within a walkable zone, it accepts the change in X.
3. Then, starting from that new coordinate, it attempts movement along the Z axis independently and evaluates again.
4. Result: If you collide head-on with a diagonal wall, the system allows you to slide along it instead of stopping.

3.9.2 Temporal Interpolation Algorithm (*Keyframes*)

It calculates exactly where an animated object should be at a given moment in time.

1. The playback advances time according to the frames per second.
2. It checks the keyframe track to find the two time markers between which the current moment falls.
3. It calculates a normalized progress factor (between 0 and 1) based on the time distance between those two points.u
4. It applies the selected easing curve to that value of .u
5. It uses a linear blend (glm::mix) to obtain the final interpolated value that will be applied on screen.

3.9.3 Hierarchical Scene Update Algorithm

In a complex 3D model, objects are connected (parent-child relationships). If the main train moves, everything inside it must move as well.

1. The system traverses the model nodes from the root to the endpoints or child objects.
2. For each node, it takes its parent's global matrix.
3. It multiplies that matrix by the original local transformation from the file and adds the local or parent animations (animPadre * local * animLocal).
4. Result: The final position in space of the entire assembly is updated in a synchronized way.

3.9.4 Hardware and software requirements

The following table shows the minimum hardware and software requirements for running the system.

Category	Element	Value
Development	Operating System	Windows 10/11 64-bit
	Language and compiler	C++, Visual Studio 2022
	Modeling	Blender 3D
	Libraries	GLFW, GLAD, GLM, Assimp, stb_image
	Video editing	Canva (demonstration video)
	Packaging	InstallForge (setup.exe)
Execution (Client/user)	Graphics	OpenGL 3.3-compatible GPU
	RAM	8 GB
	Peripherals	Keyboard and mouse

	Storage	1 GB available on hard drive

Figure 8. Minimum hardware and software requirements

3.11 Model inventory and statistics

Below is the list of the 86 models created, along with their descriptions and how each one was made.

No.	Object	AI/Own/Internet	Description
1	Car (body)	Own	Refers only to the rectangular prism geometry of the car, without considering its internal elements.
2	Metro door	Own	Rectangular geometry that opens and closes.
3	Metro door glass	Own	Translucent geometry integrated into the metro door.
4	Upper metro window	Own	The dynamic part of the metro windows, since usually only the upper part can be moved.
5	Lower metro window	Own	The static part of a metro window.
6	Metro window frame	Own	Metal object surrounding the metro window.
7	Front windshield of the car	Own	Curved front glass
8	Seats	Own	Where one sits.
9	Metal seat base	Own	Support for the car seat.
10	Handrails	Own	Where one holds on. Some were made, but they are not the ones found on the metro.
11	VIP handrail	Own	Handrail for the VIP seat, the one alone in the corner.
12	Interior door handle	Own	Located inside, to one side of where the metro doors open and close.
13	Metro lamp cover	Own	Object covering the lamp inside a metro car; lighting will be added to it in OpenGL.
14	Metro fan	Own	Object located internally at the top of the car.

15	Mexico City Metro lines map	Own	Map found inside the car and also at the station entrance.
16	Line 3 map	Own	Map showing all the stations on Line 3
17	“Emergency lever” sign	Own	Sign for the emergency lever.
18	Emergency lever	Own	Red lever.
19	Emergency lever box	Own	Support for the emergency lever
20	“Reserved seat” sign	Own	Sign identifying a reserved seat.
21	Cab (body)	Own	Refers only to the rectangular prism geometry of the cab, without considering its internal elements.
22	Speakers	Own	Geometry representing a speaker
23	Station	Own	Complete geometry of the station, understood as a large box containing all the other objects inside it.
24	Fire extinguisher	Own	Fire extinguisher
25	Station name sign	Own	Plane displaying the station name
26	"Indios Verdes" direction sign	Own	Plane displaying the station connection.
27	Women-only area sign	Own	Plane displaying "Exclusive area for women and children".
28	"Terminal Universidad" sign	Own	Sign displaying the name of the terminal.
29	"Next departure" sign	Own	Plane indicating the side from which the next metro train will depart.
30	"Next departure" alarm	Own	The alarm activated to inform the user which side the next train will depart from.
31	"Elevator" sign	Own	Plane indicating the existence of the elevator.
32	"Platforms" sign	Own	Sign indicating the direction in which the trains are located
33	"Do not enter" sign	Own	Sign placed next to the Platforms sign.

34	"Exit" sign	Own	Plane indicating the exit.
35	"Evacuation route" sign	Own	Sign indicating "evacuation route" with an arrow
36	"No smoking" sign	Own	Sign indicating that smoking is prohibited.
37	Turnstile base	Own	Base of the turnstile.
38	Turnstile doors	Own	Dynamic doors located on the turnstile.
39	Card reader***	Own	Part of the turnstile where the card is held close to gain access to the metro.
40	Metal barriers	Own	
41	Turnstile indicator light	Own	The indicator marking access through the turnstile
42	Card recharge machine	Own	Object used to recharge the metro card.
43	Candy stands - cylinder	Own	Cylinder representing a candy stand.
44	Candy stand - cube	Own	Represents an enclosed room where candy is sold.
45	Ticket booths	AI	Object representing a ticket booth for recharging cards.
46	Ticket-booth glass	AI	Translucent panel representing the glass of the ticket booths.
47	Fixed stairs	Own	Object representing a staircase.
48	Fixed stair railing	Own	Object representing the railing of a fixed staircase.
49	Elevators	Own	Cube representing the elevator.
50	Security cameras***	Own	Object representing security cameras inside the station.
51	Gravel	Own	Plane representing the gravel by the rail. A cube was made but it should be a plane.
52	Track signal	Own	Object representing a signal light.
53	Steel rails	Own	Object representing the metro rails.

54	Exclusive-area cone	Own	Cones separating the women-only area.
55	Steel fence	Own	Object simulating the separation between the metro platform and the street
56	Wooden planks	AI	Rectangular object located between the steel rails of the tracks.
57	Platform tile floor	Own	Geometry and texture of the tile on the main platform floor
58	Adult person	Internet	Female character simulating a metro user
59	Child person	Internet	Male character simulating a metro user
60	Exit 1 mural	Own	Mural located at the metro exit on the men's side
61	Metal mural protection	Own	A barrier with a long horizontal cylinder and several vertical support cylinders, all metal
62	Metal plaque with mural information	Own	Plaque containing more detailed information about a mural
63	Ticket-office mural	AI	Mural located at the station's ticket offices
64	Station lamps	Own	Unlike the metro, where only a cover is visible, the station lamps are not covered, so their geometry can be seen
65	Structural column	Own	Stone columns located inside the station
66	Exterior platform column	Own	Diagonal columns visible from the platform
67	Structural beam	Own	Horizontal bars attached to the ceiling
68	Platform structure	Own	Base structure of the platform
69	Dome	Own	Part of the roof of the metro station.
70	Yellow floor strip	Own	Yellow strip used by visually impaired people.
71	Train wheel	Own	Rubber train wheel. 1 m radius
72	Cybercenter room	AI	Room housing various electronic devices
73	Cybercenter and station manager's	AI	Door to enter the cybercenter

	door		
74	Cybercenter computer	Own	PC equipment placed on a table
75	Cybercenter computer desk	Own	Support for the PCs
76	Cybercenter attendant's desk	AI	Desk for the cybercenter attendant.
77	Cybercenter windows	AI	Cybercenter window
78	Car coupling piece	Own	What joins two cars so they do not separate.
79	Steel plaque engraved with FI	Own	Plaques containing the coats of arms of the UNAM schools
80	Station metal doors	Own	Metal mesh doors used to close the station.
81	Bicycle parking	Own	Place where a bicycle can be left, located inside the station.
82	Stairlift (stair wheelchair support)	AI	Object used so that people in wheelchairs can go up or down the stairs
83	Screens	Own	Screens.
84	Upper part of stairs	Own	Upper cube of the staircase
85	Adult man.	Internet	Adult male person to interact with the environment.
86	Adult woman	Internet	Adult female person to interact with the environment,

Figure 9. List of the 86 models used for the simulation

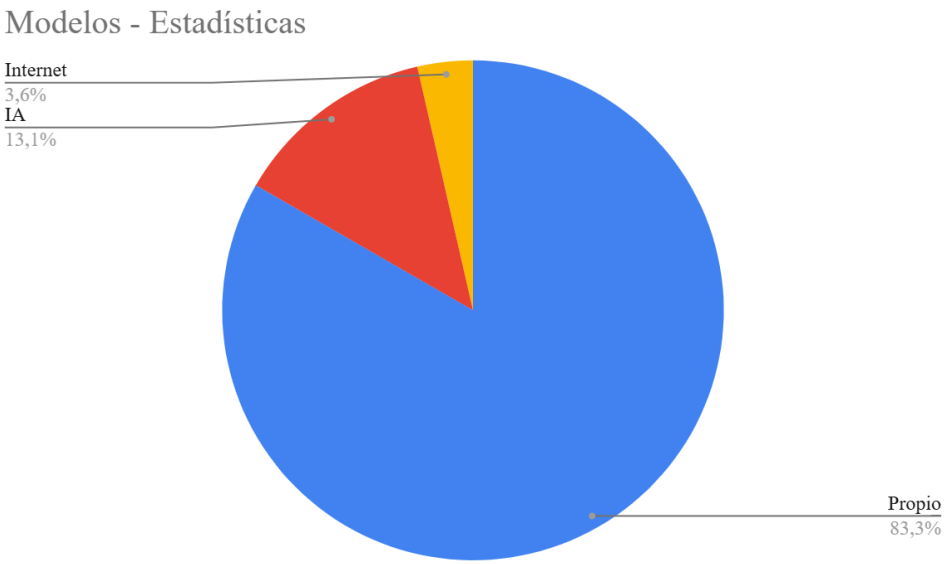


Figure 10. Chart showing the distribution of models made with AI, obtained from the INTERNET, and created by the team

4. Experiments

Experiment 4.1. Tests of exporting elements from Blender to OpenGL and displaying materials and textures

During the initial tests to determine whether Blender models were correctly exported to the OpenGL graphical environment, we encountered some difficulties related to how the materials and colors were displayed. At first, we ran export tests in .fbx format. However, when we viewed them in OpenGL, we noticed that some of the surfaces appeared darker than we expected.

Therefore, we made modifications to the mesh organization in Blender. In particular, it was necessary to separate some geometries that were part of the same object and assign their respective materials individually. We then exported the models again and ran visualization tests in OpenGL.

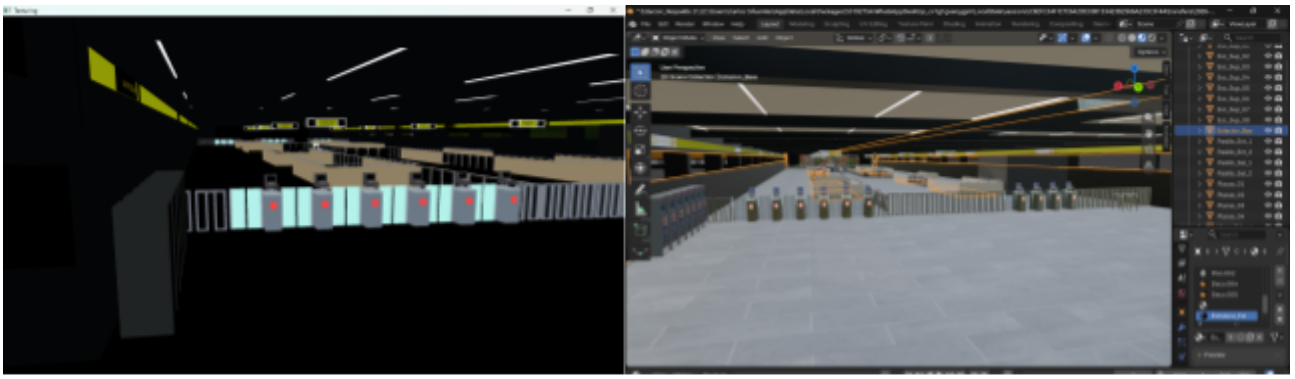


Figure 11. First visualization of the upper-floor scene when imported into OpenGL, compared with the Blender scene from a similar angle

With these changes, we were able to see how the model looked when imported into OpenGL. This experiment allowed us to identify the importance of each mesh and the way materials were assigned to them.

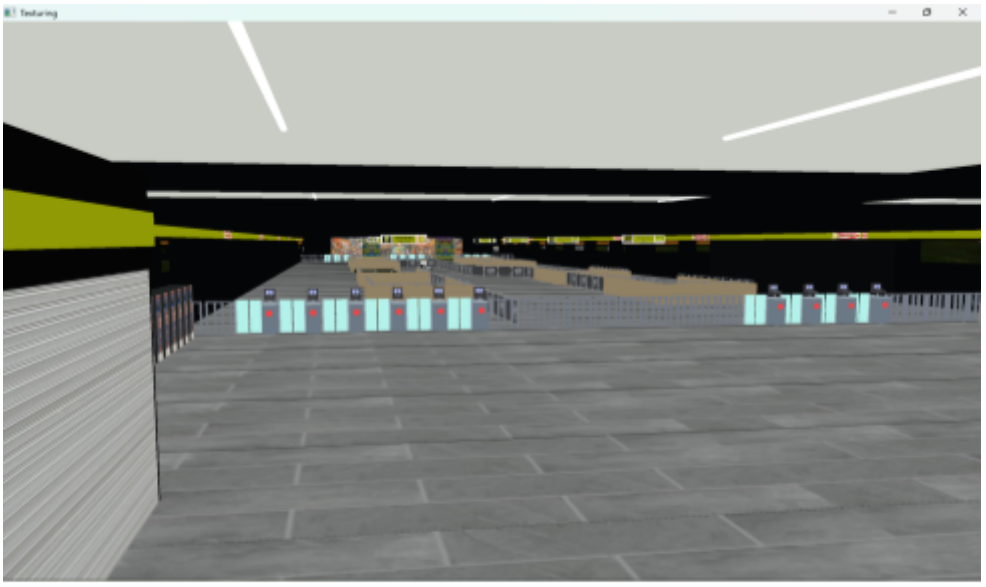


Figure 12. Scene reloaded in OpenGL with adjustments to texture export

It should be noted that lighting models had not yet been implemented in this second figure, so only the base material can be seen, without reflections or specular highlights yet.

Experiment 4.2. Tests of hierarchical organization and duplication of the cars.

During the construction of the train, different tests had to be performed to organize the components of each of the 9 cars using hierarchical structures in Blender. The objective was to group the elements that make up a car so that they could be manipulated together, without losing the ability to transform certain components independently.

In the first tests, empty objects were used as the root elements of the hierarchies. However, when we tried to duplicate one of these elements, we observed that only a copy of the root object was created, while the associated geometries could not be duplicated in the way we expected. To solve this, we reviewed the organization of the objects within their respective collections and adjusted the hierarchical relationships among the components.

Subsequently, new tests were performed to duplicate and verify that each car retained its geometric elements and could move as a single element. As a result of this experiment, it was possible to organize the different cars of the train, including the 8 simplified cars and the interactive car. The movement tests allowed us to verify that the components of each car responded to transformations applied to the root elements.



Figure 13. Empties of each car to maintain the hierarchical structure

Experiment 4.3. Transparency tests on image textures

During the process of texturing some geometric models, tests were performed so that we could incorporate images with transparent backgrounds, especially in the signs and signage used for the station. In the first tests, we observed that when certain images were applied to object surfaces in Blender, the backgrounds that were supposed to be transparent appeared black, negatively affecting their appearance.

To identify the problem, we reviewed the image we used and verified that it had an alpha channel, which makes it possible to store transparency information. However, even though the transparent regions were displayed correctly in the image, they still appeared black on the surface. This experiment helped us identify that having an image with an alpha channel does not guarantee its correct representation on a surface; it is also necessary to consider how a material is configured and how transparency is handled.

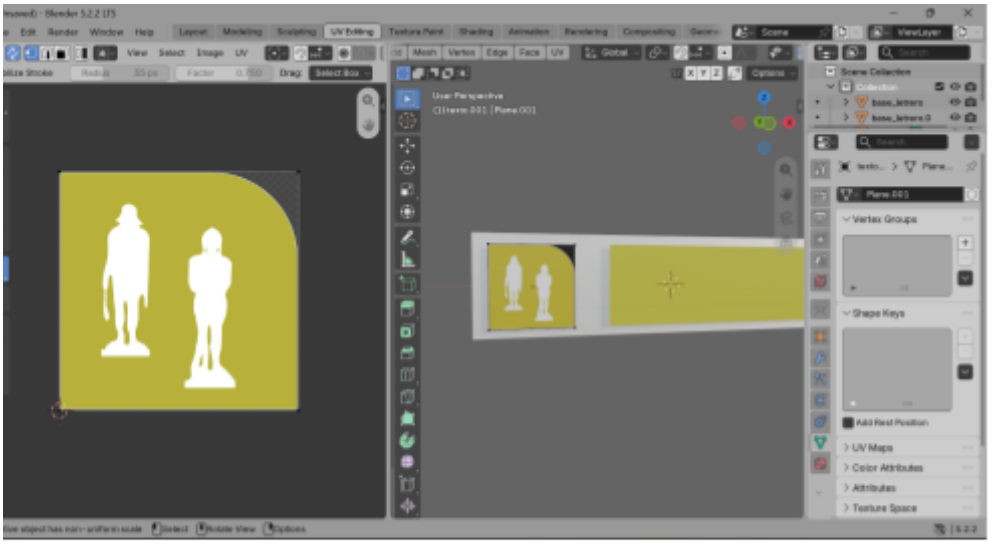


Figure 14. Failure in alpha channel assignment

Experiment 4.4. Test of importing three-dimensional models using Assimp.

Initially, we had problems related to importing .fbx files with the Assimp library, because the first tests were performed using the complete station model, but a model-loading error occurred during execution, preventing the model from being properly displayed in the graphical environment. To identify the problem, we had to perform different checks, including reviewing the path to the .fbx file and verifying its existence in the corresponding directory.

Although the file was located at the specified path, the program continued to display an error message generated by Assimp. This experiment allowed us to identify some problems during the transfer of models between Blender and OpenGL, showing that having the files in the correct location was not enough; they also had to be correctly interpreted by the libraries responsible for processing the geometries.

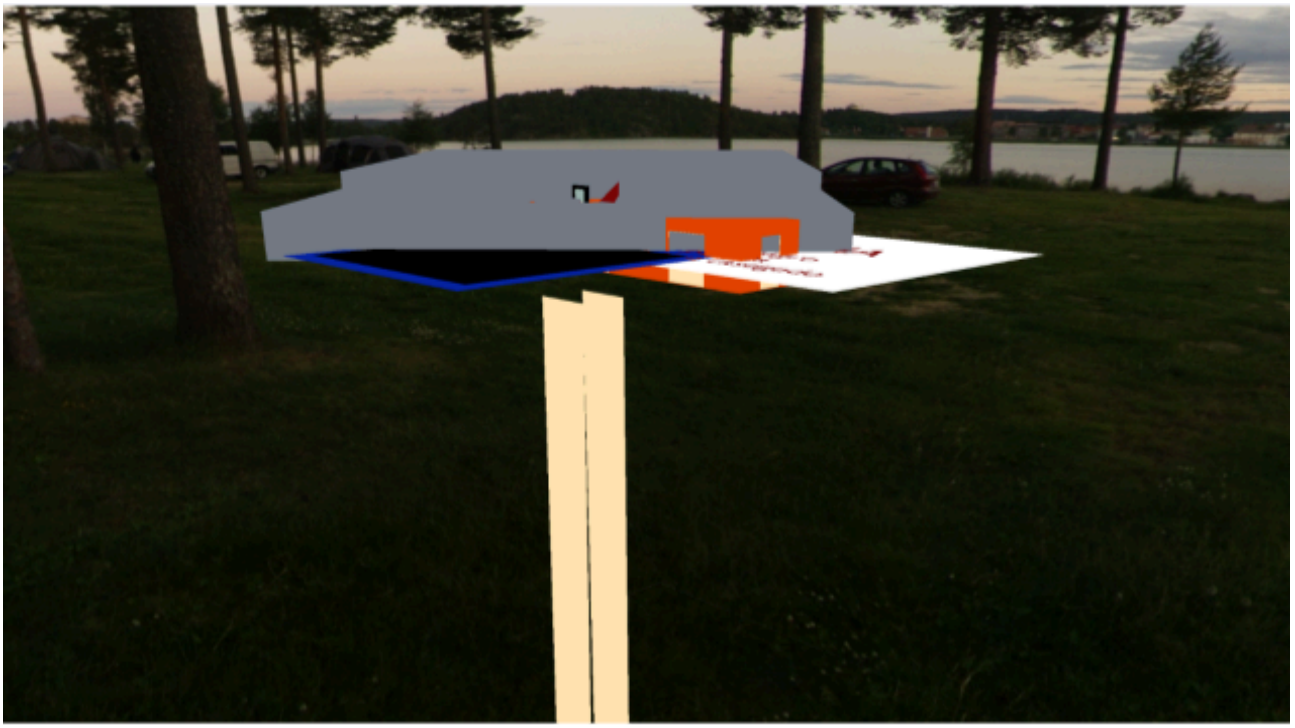


Figure 15. First import of the train model into OpenGL; incorrect interpretation of the geometries is observed.

5. Results

5.1 Final results

The graphical simulation of the station was successfully displayed in C++ and OpenGL. The environment allows the user to navigate smoothly through the first-person camera system, visually interacting with the geometric models, realistic textures, and light sources calculated in real time through GLSL shaders. The screenshots corresponding to the three constructed scenes are presented below:

Final scenes in Blender:

The following shows the scene in Blender of the metro car and its interior:



Figure 16. Final result in Blender of the exterior of the cars

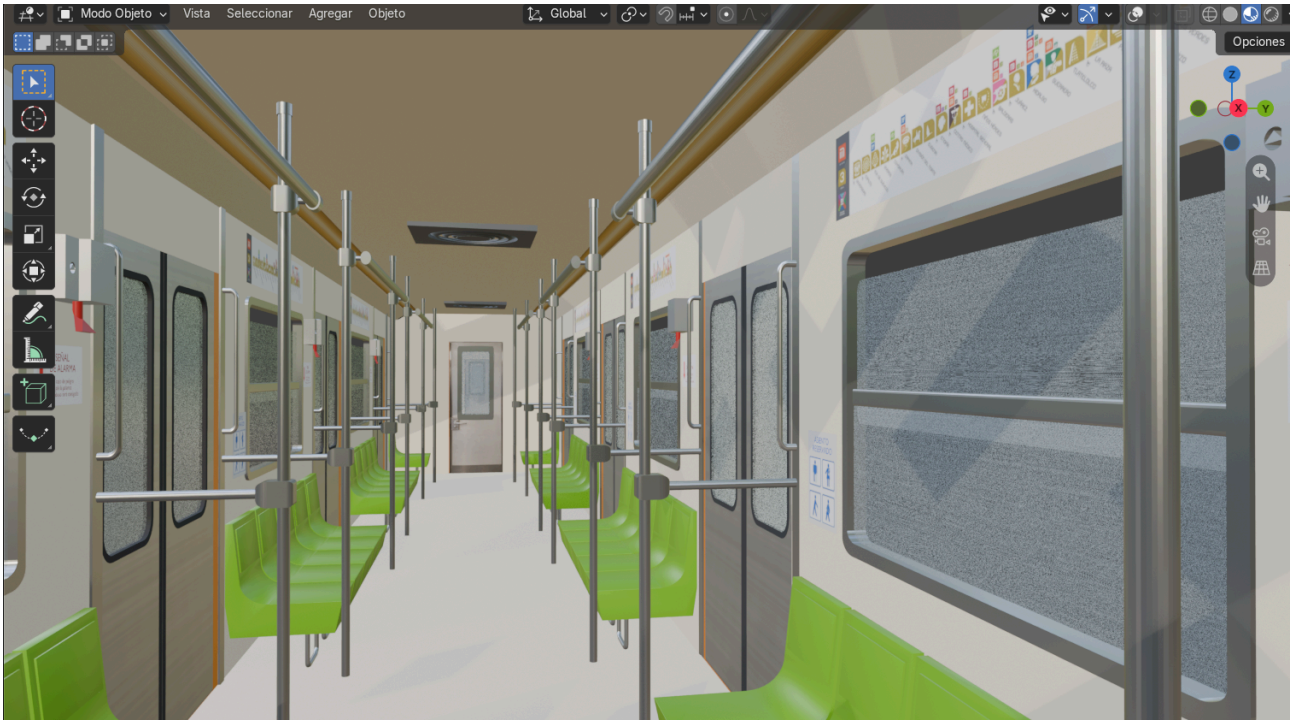


Figure 17. Final result in Blender of the interior of the metro

The following shows the scene in Blender of the platform:



Figure 18. Final result in Blender of the platform

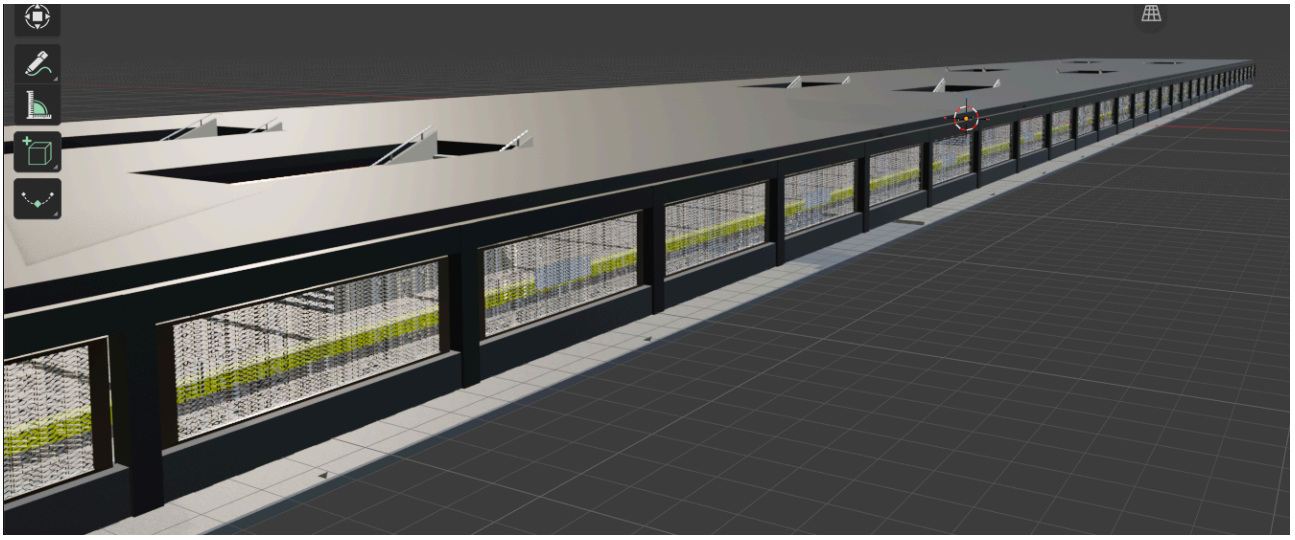


Figure 19. Final result in Blender of the platform (exterior)

The following shows the scene in Blender of the upper part of the station:



Figure 20. Final result in Blender of the upper part (interior)

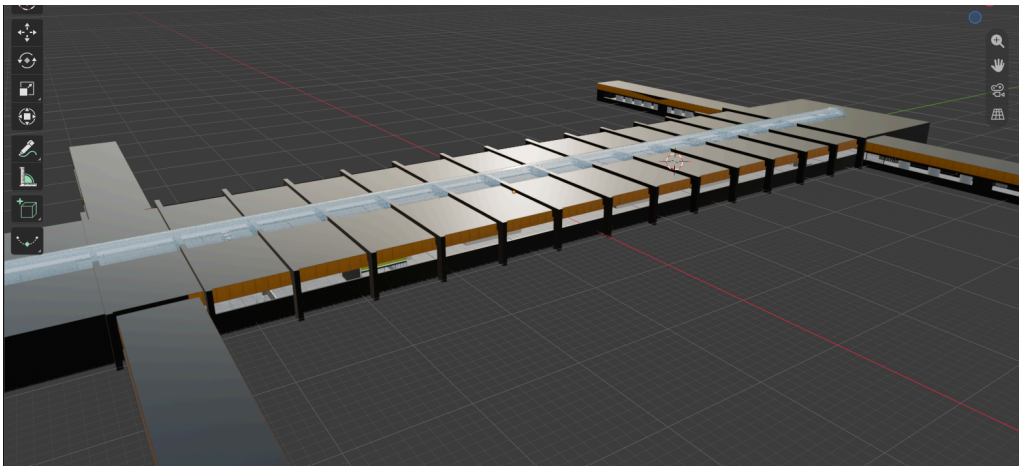


Figure 21. Final result in Blender of the upper part (exterior)

Final scenes in OpenGL:

The following shows the final scene in OpenGL of the interior of the metro car:



Figure 22. Final result in OpenGL of the interior of the car with users

The following shows the final scene in OpenGL of the platform:

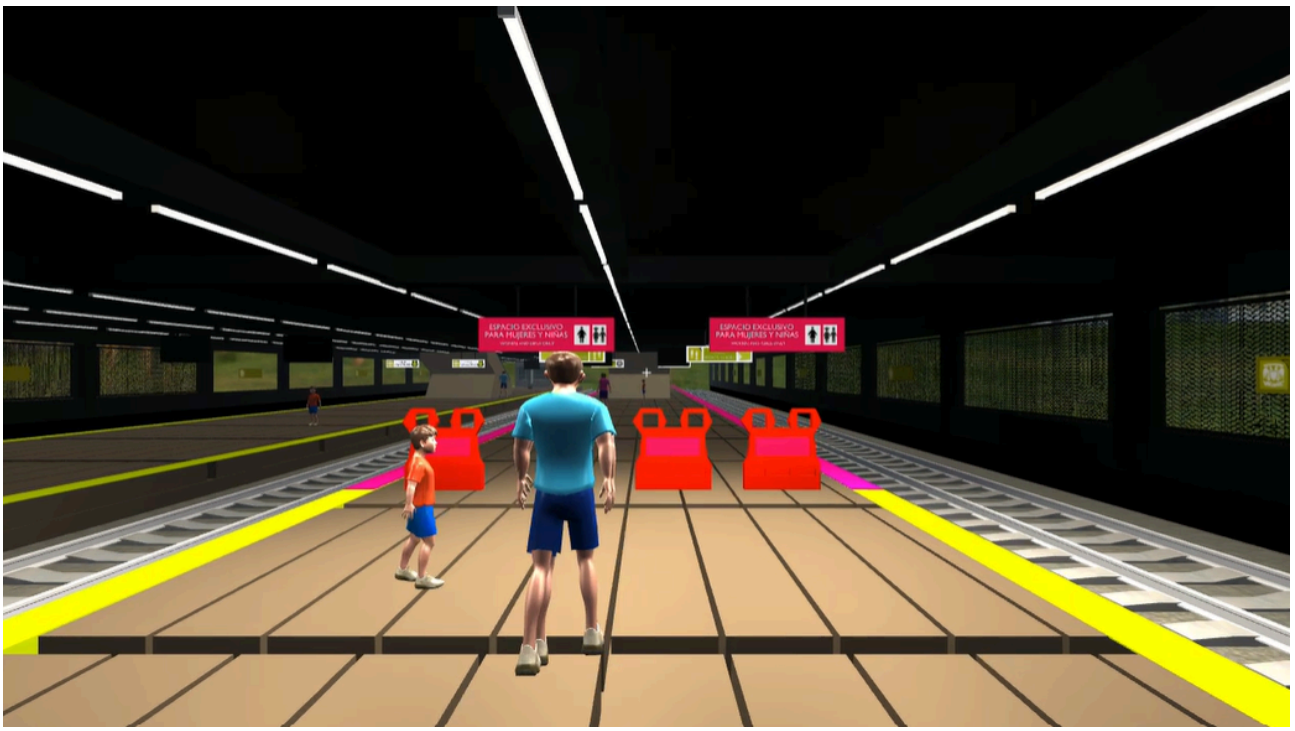


Figure 22. Final result in OpenGL of the platforms



Figure 24. Final result in OpenGL of the platforms with the metro train

The following shows the final scene in OpenGL of the upper part of the station:



Figure 25. Final result in OpenGL of the upper part (turnstiles)



Figure 26. Final result in OpenGL of the upper part

5.2 Advantages and disadvantages of the methodology applied

Among the advantages found in this project is that, by developing the system from scratch in C++ and OpenGL instead of using something more commercial such as Unity, we gained a much better understanding of the concepts, in addition to having absolute control over vertex transformations, shader compilation, and modeling calculations and everything these involve (geometric models, texturing, lighting, etc.).

Another notable advantage of the methodology used, specifically working separately on the independent scenes (entrance, platform, and train), was that it facilitated collaborative work both in modeling in Blender and in writing the code. It also improved performance on the computers so that the system did not take as long to compile and therefore did not require a very complex computing system.

On the other hand, the use of Empty nodes and parent-child relationships allowed us to manipulate the train and complex structures as blocks without losing the independence of their dynamic components. Finally, another advantage was that the import pipeline with Assimp and the extraction of packed textures guaranteed that the elements remained intact.

The methodology used also presented several disadvantages and significant challenges during development. Transferring models from Blender to OpenGL was complicated, as there were problems with transparency in the signs and signage, the texture of the car seats did not look right at first, and some surfaces required manual mesh corrections so that the model would not be too heavy

and could be displayed in OpenGL. This was the case with the platform railings, which had too many triangles and vertices and had to be optimized, although some of their shape was lost in that optimization.

Likewise, we had to program elements ourselves that are normally already integrated, such as the camera control system, movements, and basic interaction logic. This meant investing a great deal of time in debugging errors and writing code.

Finally, by loading so many detailed elements individually without implementing advanced optimization techniques (such as Instanced Rendering for repetitive objects), the system could end up requiring more resources than were available, which represented a major technical challenge for our computers.

5.3 Discussion comparing with other existing developments

When we compare our project with other simulators, such as *Train Sim World* or *OpenBVE*, the main difference lies in the approach we wanted to take. These simulators are designed to include things such as the operator's cab, controlling the brakes, accelerating the train, and ensuring that the schedules and tracks are perfect. In contrast, we decided to put the passenger at the center: someone who walks toward the entrance, passes through the turnstiles, looks for the ticket booths, goes down to the platforms, and gets onto the car. Rather than a driving game, ours is a simulation designed from the user's perspective to explore the station space and recognize its areas and signage, especially the areas reserved for older adults or people with disabilities.

On the other hand, in virtual reality studies about how people move around the metro (such as studies carried out in other countries on signage and passenger flow), we realized that almost everything focuses on transportation networks very different from our reality. What makes our project different is that it is contextualized here in Mexico City. We managed to recreate a station as representative as Universidad on Line 3 in considerable detail, respecting the elements that form part of our daily identity as students (the murals, signage, colors, and actual layout of the platforms).

6. Conclusions

Medel Piña Ángel De Jesús

With this project, we were able to use the techniques covered in the Computer Graphics laboratory, from 3D modeling in Blender (with textures, colors, transparency, etc.) to code implementation for OpenGL. The most complicated part was implementing the Blender models in OpenGL because several procedures were necessary for the textures and the model to be exported correctly.

At the same time, we realized that how we modeled had a major influence on the performance of the final program, because if one of the models was too heavy, it made the program run very slowly. Animations were also a challenge when making them work together during code implementation. The only things left to implement were shadows, sound, and a more realistic metro journey.

Oliva Santiago Vania Nayeli

The development of this project allowed us to apply the knowledge we acquired throughout the theoretical and practical course. We achieved this by creating a program that could display a three-dimensional, interactive scene of Universidad station on Line 3 of the Mexico City Metro. Throughout the project, we came to understand the importance of combining the modeling work in Blender, which was our software for this task, with transformations, lighting, texturing, and animations to construct virtual scenes that can be explored by the user. During development, we faced several challenges, such as importing models, displaying materials correctly, implementing animations, etc.

Sánchez Luján César Ricardo

Throughout this project, I was able to learn, contribute, and develop new skills, both those covered in my laboratory, which in my case was different from my classmates' because, although the same concepts were covered, they were implemented differently in code. However, it was also easy to integrate what we learned in theory, because the concepts of lighting, modeling, texturing, and animation are topics that were complemented throughout the course and put into practice in my laboratory. Another important thing I developed was the implementation of modeling in Blender, which was very interactive and easy-to-use software; along with that, developing each of the models was something fun to do for the project.

Segura Cedeño Luisa María

During the development of this project, we managed to integrate all the concepts covered in the laboratory, such as 3D modeling, texturing, animation, among others. We managed to build a 3D system of Universidad station on Line 3 of the Mexico City Metro, with most of the models being original and less than 15% external. The use of C++ and OpenGL allowed us absolute control over the pipeline, texture management, hierarchy, and lighting application.

The implementation of navigation systems and animations, together with all the modeling, created a realistic and fluid environment that allowed us to meet the project's objective: people unfamiliar with the metro could interact with the system to learn about its environments and exclusive areas and feel encouraged to use the metro if they were tourists.

Finally, it is possible to conclude that the project addresses a real and current problem of urban mobility in Mexico City. This simulation serves as an educational tool for accessibility and spatial analysis, demonstrating the value of human-computer interaction in solving problems that concern the entire community.

Sifuentes Cordero Carlos Emir

I believe that completing this project allowed me to reinforce the knowledge I acquired throughout the course, especially regarding the handling of different geometric transformations, hierarchical modeling, texturing with different materials, and animations within a 3D scene. I believe that one of the most important things was understanding how different tools are integrated, particularly Blender for modeling, C++ for programming, OpenGL as an API, and various libraries.

In addition, during development, one of the concepts that particularly surprised me was the importance of organizing models, parent-child relationships, and, also very importantly, using the appropriate shaders to represent the scene as we wanted it to look. I believe that the greatest challenge we faced as a team was when we began exporting all our models to OpenGL, because at that stage we had to find different solutions to make everything look the way we wanted. Finally, I believe that the main objectives of the project were achieved, as we were able to implement a program with all the concepts and many of the techniques we had been learning throughout the course.

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BOY CHARACTER | CGTRader. (n.d.). CGTrader. https://www.cgtrader.com/items/4511711/download-page

9. Link to the demonstration video

https://canva.link/1icw14ybyh9o2na

Appendix on AI use: prompt log

Improvement of the wording of the Introduction

Appendix: A

Tool and model: Claude, Claude Sonnet 5.5

Company: Anthropic

Chat date: November 3, 2026

Chat title: Improvement of the wording of the Introduction

Section of the paper: Introduction

Shared link: https://claude.ai/share/1c9e1d7a-63b8-4a13-b1db-e1c4d0252a5b

Full reference:

> Anthropic. (2026, November 3). [*https://claude.ai/share/1c9e1d7a-63b8-4a13-b1db-e1c4d0252a5b*] [Generative AI chat]. Claude Sonnet 5.5. https://claude.com/es/product/overview

Creation of a script to recover textures

Appendix: B

Tool and model: ChatGPT, by OpenAI.

Company: OpenIA

Chat date: November 1, 2026

Chat title: Creation of a script to recover textures

Section of the paper: Practical methodology – 3.5. Materials, colors, and texturing.

Shared link:

https://chatgpt.com/g/g-p-6ab83cb9f72c8191845114c2684ef8fd-proyecto-computacion-grafica/shared/c/6aee32f4-8510-83e8-ac71-820dc97ca267?owner_user_id=user-20Z0i4nyI1CrhUSMmcV1kY6a

Full reference:

> OpenIA. (2026, November 1). *Recovery and preparation of textures in Blender* [Generative AI chat]. ChatGPT. https://chatgpt.com/

Generation of the stairlift model and front texture

Appendix: C

Tool and model: ChatGPT, GPT-6

Company: OpenAI

Chat date: November 2, 2026

Chat title: Generation of the stairlift model and front texture

Section of the paper: Modeling and texturing of 3D objects

Shared link:
https://chatgpt.com/g/g-p-6ab83cb9f72c8191845114c2684ef8fd-proyecto-computacion-grafica/shared/c/6ae90377-aa64-83e8-93d0-31d45394258e?owner_user_id=user-WRBjwmtXI88sxd2RLxIsFHut

Full reference:

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Traducir archivo
Anexo: D
Herramienta y modelo: ChatGPT, GPT-6
Empresa: OpenAI
Fecha del chat: 06 de noviembre del 2026
Título del chat: Traducir al ingles
Sección del trabajo: Todo el documento

Enlace compartido:

https://chatgpt.com/g/g-p-6ab83cb9f72c8191845114c2684ef8fd-proyecto-computacion-grafica/shared/c/6aee86e3-ac30-83e8-b7cd-ef0b9b491513?owner_user_id=user-20Z0i4nyI1CrhUSMmcV1kY6a

Referencia completa:

Traducir al inglés. (n.d.). https://chatgpt.com/g/g-p-6ab83cb9f72c8191845114c2684ef8fd-proyecto-computacion-grafica/c/6aee86e3-ac30-83e8-b7cd-ef0b9b491513

Prompt Log:

#	Prompt (exact text)	What I used it for	What I did with the response
1	"Improve the wording of this text, without changing the structure or the information content, but only correcting punctuation, grammar errors,	Improve the wording of the text to make it more academic	I applied the suggested correction and changed/removed some words

	etc.”		
2	The texture images are only in the Blender files, and when we worked on them in Blender we configured the option External Data -> Pack Resources. After that we exported the scene to OpenGL in .fbx format by pressing Ctrl+A to apply “All Transforms.” Can you give me a script to solve this without having to go texture by texture?	Creation of a script to recover textures	The script was implemented in Blender to recover textures in an .fbm for correct visualization in OpenGL.
3	“Give me the script to model this stairlift. Focus only on the stairlift, try not to use overly complex geometries, I am not looking for too much detail.”	Creation of the stairlift model	I added the model made with AI
4	“Traducelo a ingles, no le cambies nada”	Traducir el archivo	Se añadió a la carpeta como entrega